

Important: These are benchmark questions, not legal advice. When you build your evaluation set, replace the “Expected Answer” with the wording and citations from the exact regulation or safety manual in your corpus (e.g., OISD-105, OISD-192, OSHWC Code, company HSE manual).

ID	Question	Expected Answer
Q 2 1	What is the purpose of a Permit to Work (PTW) system?	To control hazardous work through formal authorization, risk assessment, communication, and verification of safety precautions before work begins.
Q 2 2	Who is normally authorized to issue a work permit?	A competent and authorized person designated by the employer or site management.
Q 2 3	Can work begin before a permit is approved?	No. Work should only begin after the permit is authorized and all specified precautions are verified.
Q 2 4	When should a work permit be suspended or cancelled?	When conditions change, work stops for an extended period, hazards increase, or permit conditions are violated.
Q 2 5	What information should appear on a work permit?	Job description, location, hazards, control measures, PPE, validity period, authorizations, and emergency precautions.
Q 2 6	What is a Hot Work Permit used for?	Activities involving flames, sparks, or heat such as welding, cutting, brazing, or grinding.

Q 2 7	What is a Cold Work Permit used for?	Hazardous work that does not generate ignition sources but still requires controlled authorization.
Q 2 8	Why should gas testing be completed before confined space entry?	To confirm that oxygen levels are safe and that toxic or flammable gases are within acceptable limits.
Q 2 9	Who should remain outside a confined space during entry?	A trained standby attendant responsible for monitoring the entry and initiating emergency procedures if required.
Q 3 0	What is the purpose of an Emergency Rescue Plan for confined spaces?	To ensure prompt and safe rescue without exposing rescuers to unnecessary hazards.
Q 3 1	What is meant by energy isolation?	Disconnecting and securing all hazardous energy sources before maintenance begins.
Q 3 2	Which energy sources may require Lockout/Tagout?	Electrical, mechanical, hydraulic, pneumatic, thermal, chemical, gravity, and stored energy.
Q 3 3	Why must stored energy be released before maintenance?	Residual energy can unexpectedly move equipment or release hazardous forces, causing injury.
Q 3 4	What should be verified before removing locks and tags?	The work is complete, guards are restored, personnel are clear, and equipment is safe to restart.

Q 3 5	What is the purpose of a Job Safety Analysis (JSA)?	To identify hazards associated with each job step and define appropriate controls before work begins.
Q 3 6	What factors are commonly considered during a risk assessment?	Hazard severity, likelihood of occurrence, exposure, and effectiveness of control measures.
Q 3 7	What is the hierarchy of controls?	Elimination, substitution, engineering controls, administrative controls, and PPE.
Q 3 8	Why is PPE considered the last line of defense?	It protects the worker after higher-level hazard controls have been applied and cannot eliminate the hazard itself.
Q 3 9	What should be inspected before using a portable ladder?	Structural integrity, feet, rungs, rails, cleanliness, and defects.
Q 4 0	When is fall protection generally required?	Whenever work is performed where there is a significant risk of falling, as defined by applicable regulations and site procedures.
Q 4 1	What should be done before excavating near underground utilities?	Identify, locate, and isolate underground services and obtain the required excavation permit.
Q 4 2	What are the main objectives of contractor safety management?	Ensure contractors comply with site safety rules, receive training, and perform work safely.

Q 4 3	Why should contractors receive site safety induction?	To understand site hazards, emergency procedures, permit systems, and safety requirements before starting work.
Q 4 4	What information should be communicated during a toolbox talk?	Job hazards, changes in work conditions, control measures, PPE requirements, and emergency actions.
Q 4 5	What is the purpose of incident investigation?	To identify immediate and root causes and implement corrective and preventive actions.
Q 4 6	Why should near misses be reported?	They provide opportunities to prevent future accidents before injuries or damage occur.
Q 4 7	What is meant by a competent person?	An individual with the necessary knowledge, training, skills, and authority to perform assigned safety responsibilities.
Q 4 8	What is the objective of periodic safety inspections?	To identify unsafe conditions and behaviors before they result in incidents.
Q 4 9	What should an emergency evacuation plan include?	Alarm procedures, escape routes, assembly points, communication methods, and accountability for personnel.
Q 5 0	What documents should be available during a safety audit?	Risk assessments, permits, training records, inspection reports, maintenance records, incident reports, and emergency plans.

Higher-Difficulty RAG Evaluation Questions

These require retrieving and combining information from multiple documents rather than quoting a single definition.

I D	Question	Expected Outcome
Q 5 1	A welding job is planned inside a storage tank. Which permits are likely required?	Hot Work Permit, Confined Space Entry Permit, gas test certificate, LOTO/isolation verification, rescue plan, and JSA.
Q 5 2	An employee discovers damaged PPE before starting work. What should they do?	Stop using the PPE, report the defect, replace it with approved PPE, and do not begin work until suitable protection is available.
Q 5 3	During maintenance, a new hazard is identified that was not included in the permit. What action should be taken?	Stop work, reassess the risk, revise or reissue the permit if necessary, and only resume after controls are implemented.
Q 5 4	Why should a permit have an expiry time?	Because hazards and site conditions can change, requiring periodic reassessment before work continues.
Q 5 5	How does a safety manual differ from a Standard Operating Procedure (SOP)?	A safety manual defines organizational safety policies and systems, while an SOP provides detailed step-by-step instructions for performing a specific task safely.

Q 5 6	Why are toolbox talks conducted daily on many worksites?	To communicate job-specific hazards, reinforce controls, discuss changes, and improve safety awareness before work starts.
Q 5 7	What is the difference between a hazard and a risk?	A hazard is a source of potential harm; risk is the likelihood and severity of that harm occurring.
Q 5 8	Why should emergency drills be conducted periodically?	To verify preparedness, test emergency procedures, identify weaknesses, and improve response effectiveness.
Q 5 9	What information is typically included in a contractor pre-qualification process?	Safety performance, training records, competency, certifications, insurance, and previous incident history.
Q 6 0	Why should safety documentation be controlled and periodically reviewed?	To ensure personnel use current procedures, comply with regulations, and incorporate lessons learned or regulatory updates.

Tip for your evaluation dataset

For each benchmark Q&A, store:

- **Question**
- **Gold answer**
- **Source document** (e.g., OISD-105, OISD-192, OSHWC Code, Company HSE Manual)
- **Section/Clause/Page**
- **Expected citations**
- **Difficulty level** (Easy, Medium, Hard)
- **Topic tag** (PTW, LOTO, Fire Safety, PPE, Confined Space, etc.)

This structure makes the dataset much more useful for objectively evaluating retrieval accuracy, answer correctness, and citation quality.